Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and frontier AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test this instead in a silicon analogue: a navigation agent where only the visual sensor varies along an axis of encoder spatial bandwidth, with architecture, task, and training held fixed. Across this axis, agents reach near-identical task performance, yet the recurrent-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the recurrent state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift rather than a total-information collapse. We frame this as an emergent *capacity allocation* between the visual encoder and the recurrent integration of GPS-sensor history. Behaviourally, memory transplants are asymmetric (rich-encoder policies fail to use bottleneck states, while bottleneck policies still drive themselves from rich-encoder ones), indicating policy-level consequences of the format differences. Probe-readability and policy reliance further dissociate as independent axes: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. The pattern connects to the invariance–disentanglement reading of the information bottleneck and to the no-remapping vs. global-remapping axis of hippocampal coding under hidden-state inference. This positions sensor structure as a training-time lever on the format of cognitive maps in artificial agents, offering a content-level alternative to training-distribution accounts of vision–language spatial-reasoning failures. More broadly, this capacity allocation generates falsifiable predictions for transformer-based world models that share the encoder–memory pipeline, and suggests that bandwidth restriction may be cognition-enabling rather than merely compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture — vision–language models with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025], with two competing diagnoses: a training-distribution gap [Chen et al., 2024], and an encoder-side argument that pretrained encoders carry markedly different geometric content depending on training objective [Tong et al., 2024, El Banani et al., 2024]. We pursue an even sharper structural reading: the encoder and the memory downstream of it are a co-trained system, and *the encoder's structure shapes what is linearly readable from the memory about space* — a content-level claim about the perception–memory interface.

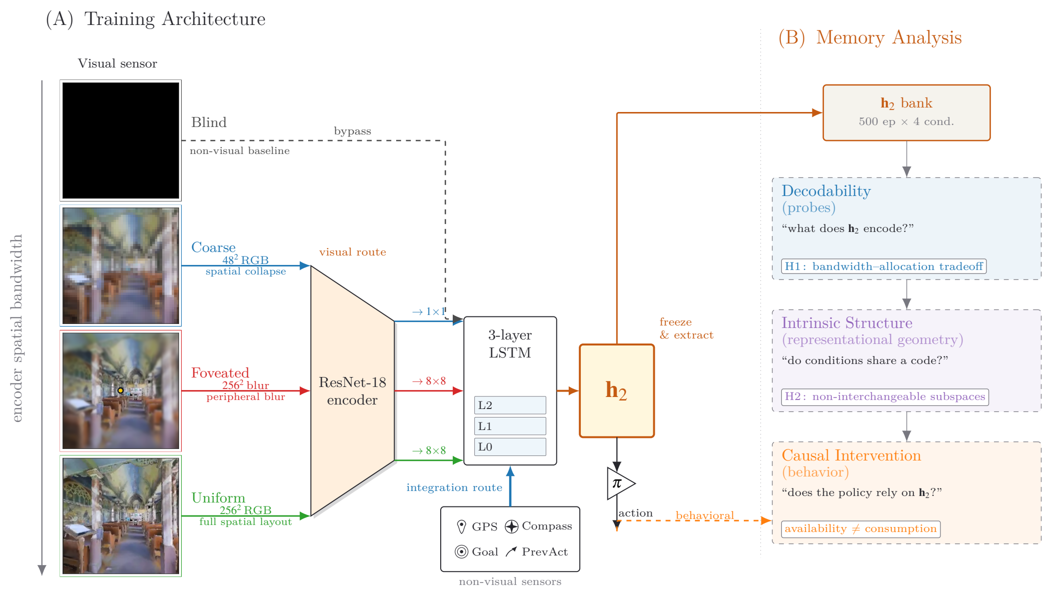

Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span an axis of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix G) adds a 5th intermediate point on the same axis. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three axes of one capacity allocation: *magnitude*, *format*, *consumption* (§2; synthesis in §4).

Cognitive neuroscience offers concrete precedent for this at the species level: primate view-cells under foveated saccades versus rodent place-cells under near-panoramic vision [Wirth et al., 2017, Rolls, 2024]; cross-modality remapping in echolocating bats whose hippocampal coding reorganises between vision and echolocation [Geva-Sagiv et al., 2016]; congenitally blind humans showing larger anterior hippocampal volume and supranormal non-visual navigation [Fortin et al., 2008]. Whether this generalises to artificial systems is hard to test in animals — their sensor ecology is fixed within species, and modern AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test it in silicon, where this design degree of freedom is available.

Cognitive-map-like representations emerge in the recurrent state of deep-RL navigation agents even with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018],[1] but the literature studies one condition at a time, leaving open how encoder structure interacts with what the memory builds. We adopt this agent family as a comparative chassis (Figure 1): identical encoder–memory–policy pipeline + task + training; varying only the visual sensor across five conditions on an axis of encoder spatial bandwidth (§2).

If the visual encoder and the recurrent integration of GPS-sensor history are competing routes for a finite recurrent-memory budget, *then* richer encoders should erode the linear-readable position signal in the recurrent state while leaving task performance intact. We observe exactly this: across the five sensor conditions, top-layer linear GPS R^2 falls monotonically with encoder spatial bandwidth while success rate stays at 93–99%. We frame this as an emergent *capacity allocation* between visual and integration routes and test it along three orthogonal axes: *magnitude* (how much linear position information lives in the recurrent state), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). Cross-method convergence across linear/MLP/MINE probes, leave-one-scene-out validation, Procrustes shape distance, 5×5 memory transplant, and shortcut-discovery tests rules out single-method artefacts; the asymmetric

[1]"Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

direction-of-transplant result establishes the format claim outside any probe framework [Orozco et al., 2025].

Contributions. (i) A 5-condition controlled silicon model isolating sensor-induced spatial-format change while holding encoder backbone, task, training, and reward fixed. (ii) The *capacity-allocation* principle: a single mechanism organising five concomitant signatures (magnitude collapse, format divergence, consumption dissociation, scene-invariance, place-vs-view-cell character) into three orthogonal axes. (iii) An asymmetric memory-transplant test (rich-encoder policies fail to use bottleneck states; bottleneck policies still drive themselves from rich-encoder ones) that mirrors hippocampal cross-modality remapping at the behavioural level. (iv) A falsifiable architecture-agnostic prediction for transformer-based navigators (§4, cf. Whittington et al. 2022). For visual intelligence, capacity allocation offers an architecture-side complement to data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures, and reframes bandwidth-restricting perception (foveation, log-polar) as potentially cognition-*enabling* rather than only compute-saving.

2 Methods

Architecture, task, and conditions (Figure 1, Row 1). PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020] across five conditions varying only the visual sensor: *blind* (no encoder); *coarse* (48×48 RGB, 1×1 encoder feature map after ResNet-18 downsampling); *foveated* (256×256 + eccentricity-dependent Gaussian blur $\sigma_{\max}$=8, 4×4 feature map); *uniform* (256×256 unblurred, 4×4); *foveated-logpolar* ($\sim 2\times2$ via log-polar resampling onto a 64×64 retinal grid, control point isolating encoder-spatial-output from blur, Appendix G). All five share an identical 3-layer LSTM (512-d) + Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal); Gibson-0+ [Xia et al., 2018] + MP3D training pool; 18 MP3D test scenes for distribution-shift probes; 250M frames each; single seed per condition (cogsci modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]; rigour from convergent multi-method evidence, §4.5); blind has an independently retrained checkpoint (Appendix A). After training, weights are frozen and per-step top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ is collected from $n = 500$ Gibson held-out deterministic-rollout episodes per condition.

Probing methodology (Figure 1, Row 2). Three method blocks read $\mathbf{h}_2$ to test three orthogonal axes of one capacity allocation:

- *Decodability* $\rightarrow$ *magnitude* axis: linear Ridge with Hewitt–Liang selectivity [Hewitt and Liang, 2019], 2-layer MLP, MINE [Belghazi et al., 2018], Skaggs spatial information [Skaggs et al., 1996]; plus step-binned, lag-k, predictive-horizon [Stachenfeld et al., 2017, Whittington et al., 2020], and excursion-forgetting variants for memory dynamics.
- *Intrinsic structure* $\rightarrow$ *format* axis: LOSO cross-validation [Sanders et al., 2020], cross-condition probe transfer, CKA [Kornblith et al., 2019], 1-NN purity, Procrustes shape distance [Williams et al., 2021], principal angles + position-direction cosines, participation ratio and TwoNN intrinsic dimension [Ansuini et al., 2019, Facco et al., 2017], eigenspectrum power-law [Stringer et al., 2019].
- *Causal/behavioural intervention* $\rightarrow$ *consumption* axis: 5×5 memory transplant + memory-init transplant; paired same-scene shortcut-discovery (Tolman 1948 in silico); GPS-sensor mid-rollout ablation[2].

Each method is introduced at point of use in §3; full protocols and hyperparameters in Appendix B. Convergence across the three axes on the same condition-level contrast is the strongest claim the methodology supports.

3 Results

3.1 Comparable navigation skill across all five conditions

Table 1 collects the headline per-condition numbers. **All five agents learned; none failed.** The four sighted conditions reach 98–99% success and blind reaches 93%; a Bug-style controller without

[2] Zero-input GPS/compass ablations push the LSTM off its training distribution; we use them only at the policy-output (success) level.

Table 1: Per-condition headline. 250M frames each (§2); blind uses an independently retrained checkpoint (Appendix A). Probe R^2: 5-fold episode-level CV mean $\pm$ std on full-length trajectories (no per-episode step cap). Sub-chance R² in rich-encoder rows reflect long-tail collapse (mid-episode $R^2 \in [0.6, 0.95]$, Figure 24b); rich-encoder GPS R^2 is statistically indistinguishable from zero (one-sample t-test: foveated $p \approx 0.58$, uniform $p \approx 0.25$; df=4). Bold marks blind/coarse rows where GPS+Compass remain decodable ($p < 0.001$) and best task performance.

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	250M	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	250M	0.853	0.989	**+0.582±0.099**	**+0.659±0.061**
Foveated	250M	**0.894**	**0.993**	+0.178±0.659	+0.503±0.142
Foveated-logpolar	250M	0.879	0.990	−0.029±0.759	+0.467±0.330
Uniform	250M	0.891	**0.993**	−1.785±2.993	−1.244±3.579

learning achieves SPL 0.07 on the same scenes — an order of magnitude below every trained agent (Appendix B). The narrow success bracket across conditions rules out a skill-gap account for any cross-condition internal difference below.

The five agents are shown different visual inputs but reach near-identical task competence on identical task and training. *They must therefore have learned different internal representations of the same task.* The open question the rest of §3 addresses is *what kind* of representation each condition learned: how the recurrent state encodes spatial position, how those encodings differ across conditions, and what this says about the encoder–memory race.

3.2 Magnitude axis: linear position code shrinks with encoder bandwidth

The magnitude-axis claim, precisely. Rich visual input *reduces the linear readability of the GPS-derived world-frame position signal at the policy-readable LSTM layer*: as encoder spatial output dimension grows, top-layer linear GPS R^2 falls monotonically across the four main conditions (Figure 2), with the foveated-logpolar control at $\sim 2{\times}2$ landing in the rich-encoder regime — a falsification of the simplest spatial-output-dimensionality reading and motivation for the spatial-feature-variety reframing (Appendix G). Goal-distance decodes at $R^2 \in [0.81, 0.90]$ in every condition, ruling out a probe-capacity reading. The contrast is specifically about whether a *linear* top-layer readout — the operation the DD-PPO policy head performs — can recover position, not whether position is encoded anywhere in $\mathbf{h}_2$. The blind result replicates [Wijmans et al., 2023]; coarse extends it to a condition with visual input but $1{\times}1$ encoder output, suggesting the shared trigger is minimal visual-feature variety per step, not absence of vision. Throughout this section "GPS code" is shorthand for "linearly decodable top-layer GPS-derived position signal".

Information allocation: linear collapse vs. MLP recovery. The format-shift-not-info-collapse claim is sharpened by three measurements (Figure 25, Appendix E.4). Linear (Ridge $\alpha = 10$) GPS R^2 varies across a ≈ 2.1-magnitude range over the five conditions ($+0.94$ blind to -1.19 uniform). A 2-layer MLP probe compresses this to $\approx$ 0.65 ($+0.99$ blind to $+0.48$ uniform), recovering $R^2 \in [0.51, 0.73]$ in rich-encoder conditions where linear is at chance; the linear–MLP gap widens monotonically with encoder bandwidth. MINE estimates of total $I(\mathbf{h}_2; \text{pos})$ shift only $\approx 1.3\times$ across the same range. Position information is approximately preserved by non-linear readout while linear-readable information collapses — a format shift, not a total-information loss. Since the DD-PPO action head is a linear projection of $\mathbf{h}_2$, "linearly decodable at L2" maps directly to "policy-accessible".

Temporal dynamics and capacity reallocation. We propose this reflects a *capacity-allocation* mechanism: the L0 GPS-sensor input integrated across time and the encoder's position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to integration; rich-encoder conditions provide a viable visual route and policy gradient reallocates capacity to it. Two within-episode and cross-training observations support this. *Within-episode* (step-binned probing, Figure 24b): rich-encoder conditions have R^2 of 0.6–0.8 in the typical mid-episode window (steps 50–200) but drop to chance in the long-tail (steps 1024+); the lag-k retrospective probe corroborates ($R^2 \geq 0.89$ for blind across $k = 0$–20, with rich-encoder profiles unstable, Appendix B). *Across training* (Figure 26): all four sighted conditions show $R^2 \approx +0.91$ at ~ 50M frames, then the three rich-encoder conditions decay along a trajectory whose timescale

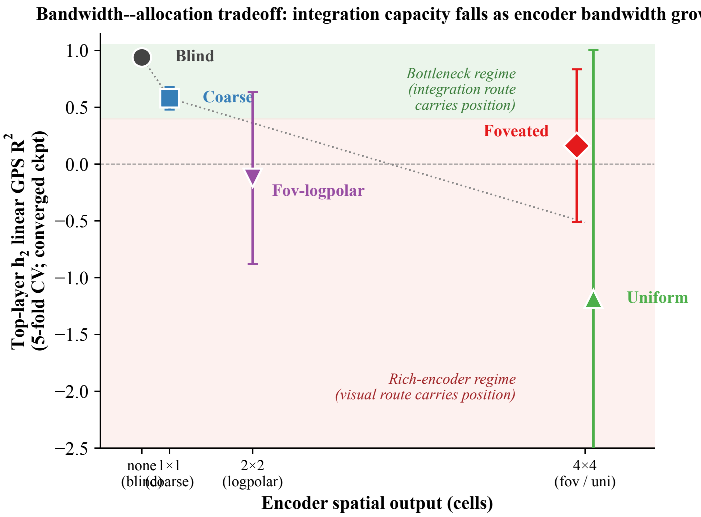

Figure 2: **The capacity-allocation principle, quantitative form.** Top-layer $\mathbf{h}_2$ linear GPS R^2 (5-fold CV mean $\pm\sigma$) plotted against encoder spatial output dimension. All sighted conditions at 250M frames; blind at 340M (Appendix A). Conditions: blind (no encoder), coarse (1×1), foveated-logpolar ($\sim 2\times2$, Appendix G), foveated (4×4 blurred), uniform (4×4). The monotonic anti-correlation is the bandwidth–allocation tradeoff: as encoder spatial bandwidth grows, less integration capacity remains in the recurrent state. Foveated and uniform sharing x but differing in y shows the design space is not 1-D — channel-level information is a second axis (foveation hybrid case discussed in §4).

tracks encoder informativeness (uniform fastest, foveated/foveated-logpolar slower) while coarse preserves $R^2 \in [0.43, 0.58]$ throughout. Compass codes follow the same pattern (Figure 26b). The pipeline-view decomposition (Figure 24c) localises the divergence: the encoder is at chance for all sighted conditions; the L0 jump is the GPS-sensor embedding entering the LSTM; the bottleneck-vs-rich-encoder split emerges specifically at L2.[3]

Population coding and perturbation evidence. At the population level (Figure 27), per-unit Skaggs spatial information [Skaggs et al., 1996] is essentially condition-flat (1.16–1.39 bits, 0.23-bit range) but place-unit count ($>$ 1 bit) reverses the gradient (blind 259/scene vs. sighted 284–303/scene): blind concentrates spatial information into fewer, more-informative units. Per-unit regression on the top-20 Skaggs-peaked units sharpens the population reading: in all sighted conditions peaked units are better explained by trajectory-phase features (distance-to-goal, step-in-episode) than by raw position ($R^2_{\text{traj}} > R^2_{\text{pos}}$ by $+0.05$ to $+0.15$); for blind the direction reverses ($R^2_{\text{pos}} = 0.14 > R^2_{\text{traj}} = 0.09$). Rich-encoder peaked units therefore encode position-correlated features (visual landmarks, trajectory phase) consistent with view-cell-like binding [Rolls, 2024]; blind's broader, lower-peak distribution carries the linearly-decodable position code. *Predictive-horizon probes* [Stachenfeld et al., 2017, Whittington et al., 2020] (Appendix E.3) and an *excursion-forgetting test* converge on the same dichotomy. Blind sustains $R^2 \in [0.90, 0.94]$ from $k = 0$ to $k = 20$ on future-position decoding (the SR-style cognitive-map signature), and $+0.79$ at $k = 50$; coarse mirrors at $R^2 \in [0.34, 0.57]$; rich-encoder conditions show no predictive horizon. Under a 25-step random-action mid-episode excursion (Figure 21), blind's position-decoding fragments most ($\Delta\text{MAE} = +0.381$ vs. sighted 0.13–0.21): *the more a condition's* $\mathbf{h}_t$ *integrates the action stream, the more it fragments under perturbation* — the inverse of a passthrough-style hidden state. The reverse direction of forgetting magnitude, vs. a prior reading that placed blind on the most-robust end, follows from the post-retrain re-analysis with hp-consistent ckpt.25; we treat the "fragility-tracks-integration" interpretation as best-fitting the new data but as an interpretive call rather than a directly tested causal claim.

[3] The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\to$ $+0.35$ — awaits multi-seed replication. The MLP result alone cannot exclude exploitation of position-correlated features (Appendix C). A representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

3.3 Format axis (within-condition): scene-invariance distinguishes bottleneck from rich encoders

The linear→MLP gap (Figure 25) shows position information is approximately preserved by MLP probing across conditions even when linear readability collapses — a format shift rather than a total-information collapse (MINE shows a more modest $\approx 1.3\times$ decrease in total $I(\mathbf{h}_2; \mathrm{pos})$; Appendix E.4). What changes *format* between bottleneck and rich-encoder conditions? A leave-one-scene-out (LOSO) cross-validation localises the answer: for each held-out Gibson scene, fit Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene (Figure 22). Blind achieves median LOSO $R^2 = 0.90$ with 2% of held-out scenes giving negative R^2; coarse drops to median 0.17 (48% scenes negative); foveated 0.28 (38%); uniform 0.15 (44%). The bottleneck-vs-rich-encoder rank ordering is preserved across the cross-training trajectory (Figure 26), confirming the LOSO contrast is not specific to the 250M endpoint.

The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (preserved, per the MLP probe) but the *scene-invariance* of the linear readout direction. A bottlenecked encoder has no scene-specific features to condition on; the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. A rich encoder lets the LSTM use scene-specific visual features to encode position; different scenes induce different position-directions in $\mathbf{h}_2$, and a single linear probe cannot capture this scene-conditional encoding while an MLP can by switching its readout on visual context. This re-states capacity allocation in IB-invariance terms [Achille and Soatto, 2018]: bottleneck-induced encoding is scene-invariant (a single linear axis serves all scenes), while rich-encoder encoding is scene-conditional (the linear axis is scene-dependent and only an MLP can switch readouts on visual context).

3.4 Format axis (across-condition): conditions allocate to non-interchangeable subspaces

If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition's allocation cannot read another's, and a memory transplant should fail. The 5×5 memory-transplant matrix at midpoint t=30 (Figure 23, right) shows a marked direction-asymmetry: when a bottleneck condition (blind) provides the donor and a rich-encoder condition is the recipient, recipient SPL drops by 0.17–0.33 (mean -0.23); reversed (rich-encoder donor, bottleneck recipient), recipient SPL is preserved (mean $\Delta \approx 0$, range -0.07 to $+0.06$). *Rich-encoder policies cannot use bottleneck-style hidden states; bottleneck policies can drive themselves from rich-encoder ones.* Recipient sensitivity scales with encoder bandwidth (uniform most brittle, blind least), and condition pairs that look indistinguishable under linear CKA still differ by 0.01–0.38 SPL under transplant.[4] Cross-condition probe transfer (Figure 23, left) mirrors the asymmetry: blind→rich-encoder transfer is mild ($R^2 \in [-0.28, -0.15]$), rich-encoder→blind catastrophic ($R^2 \in [-6, -1.2]$). 1-NN purity is 1.000 (chance 0.20) on $10\,000\times5$ pooled top-layer states; probe-overfit accounts are ruled out by 5-fold episode-level CV, self-probe $R^2 \in [0.6, 0.95]$, and Hewitt–Liang selectivity tracking each probe's R^2.

Geometric scalars (Table 2). Three methodologically distinct geometric views corroborate. *Procrustes shape distance* [Williams et al., 2021] (a true Riemannian metric on Kendall's shape space, unlike CKA): pairs in a bandwidth-allocation-aligned pattern — bottleneck pair blind–coarse and rich-encoder pair foveated–uniform are tighter than cross-regime pairs. *Participation ratio + TwoNN intrinsic dimension* [Ansuini et al., 2019]: coarse separates from the other three (PR 3.10 vs. 4.01–4.34; ID 1.64 vs. 1.77–1.85), consistent with 1×1 encoder collapse pulling down representational dimensionality. *Eigenspectrum power-law* [Stringer et al., 2019] (PCs 5–200 Stringer protocol, Appendix E.5): the four sighted conditions cluster around the Stringer optimum $\alpha = 1 + 2/d \approx 1.67$ (1.72–1.90), while blind sits well above ($\alpha = 3.85$), a sharp tail decay consistent with variance concentrated in ~ 4 dominant top PCs (top-1:top-10 ratio $\approx 31\times$, top-10 PCs capturing 95% of variance). PR distinguishes coarse's encoder-collapsed regime; α distinguishes blind's tightly-decayed tail. The Stringer interpretation is descriptive rather than mechanistic: our setting lacks the stimulus-noise structure his theory was built for.[5]

[4] The matrix excludes the 6 coarse cross-pair cells due to differing input size (48^2 vs. 256^2); the asymmetry test relies on the blind axis only. Specific cell magnitudes will move with replication.

[5] Procrustes pairing is per-episode mean (Appendix E for alternatives); ID estimators MLE and Facco-CDF agree within 0.15; eigenspectrum α fit on log-log of PCs 5–200 of mean-centred $\mathbf{h}_2$ (n=$30\,000$ subsampled steps). Single seed per condition (§4.5).

Table 2: **Representational geometry.** *Within-condition (left):* PR = participation ratio (effective dimensions); TwoNN ID-MLE and ID-CDF (intrinsic dimension); α = eigenspectrum power-law exponent (PCs 5–200, Stringer 2019; optimum $\alpha = 1 + 2/d \approx 1.67$). *Cross-condition (right):* Procrustes Riemannian angular distance θ_1 on top-PCA shapes (range $[0, \pi/2]$; larger = more distinct). Bold = column-wise minimum (PR, ID) or closest to optimum (α). Coarse separates on PR/ID (encoder collapse); blind on α (sharp tail). θ_1 pattern is bandwidth-aligned: bottleneck pair (blind–coarse) and rich-encoder pair (foveated–uniform) are tighter than cross-regime pairs. 500 paired Gibson eval episodes; single seed.

	Within condition				*Procrustes θ_1 to other condition*			
Condition	PR	ID-MLE	ID-CDF	α	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	**3.85**	–	1.10	1.23	1.22
Coarse	**3.10**	**1.76**	**1.64**	1.80	1.10	–	1.14	1.13
Foveated	**4.34**	1.89	1.77	**1.72**	1.23	1.14	–	1.08
Uniform	4.01	1.91	1.80	1.90	1.22	1.13	1.08	–

Direct subspace test and early commitment (Figures 28, 29). The capacity-allocation principle predicts mutually orthogonal subspaces of $\mathbf{h}_2$ with position encoded along non-aligned directions. Across all ten condition pairs, mean principal angles between top-K PCA subspaces ($K \in [4, 33]$ per condition covering $\sim 90\%$ variance) lie in $[69, 87]$, and pairwise cosines between Ridge-probe GPS-direction weights lie in $[-0.094, +0.102]$ — indistinguishable from random.[6] The allocation is committed *early*: at the earliest probed checkpoint ($\sim$ 50M frames), principal angles are already $[82, 89]$ and direction cosines are essentially zero — the structural divergence is a training-time decision laid down well before convergence, not a late-emerging novelty.

3.5 Consumption axis: probe-readability dissociates from policy reliance

Probe-readable position indexes *where capacity is allocated*; whether the policy actually uses it is independent. *Shortcut discovery* [Tolman, 1948] (paired same-scene-different-goal episodes; SPL drop under reset vs. persistent LSTM) reveals two off-diagonal cases against probe-readable top-layer GPS: *coarse* carries a linear GPS code yet has a low shortcut SPL drop (policy under-weights the readable GPS); *uniform* has no linear GPS yet has a high shortcut SPL drop, so the policy must read a non-spatial substitute. A direct readout test confirms it: a multinomial scene-id classifier on $\mathbf{h}_2$ achieves 95.5% ($126\times$ chance) for uniform vs. 85.5% ($100\times$) for blind, while within-scene linear position $R^2 = 0.86$ uniform vs. 0.98 blind — uniform's $\mathbf{h}_2$ encodes scene identity *relatively more* than position, the inverse of blind. Trajectory-level analysis (Figure 30) gives a candidate mechanism: only uniform's persistent agent ends up closer to the *previous* episode's goal than the new one (margin $+1.83$m, n=46); the LSTM tells uniform's policy "head to where you were going last episode", a visual / scene-history anchor. A within-condition complement comes from *memory-init transplant* (re-initialise from previous-episode end-memory vs. zero, n=150 ep/cond): all conditions show ΔSPL $\in [-0.10, -0.14]$, with blind's success dropping $0.81 \rightarrow 0.67$. The policy at t=0 expects a zero hidden state regardless of encoder type, so capacity allocation is *cross-condition* non-interchangeable *and within-condition* trajectory-bound.[7]

Behavioural validation under GPS-sensor ablation. Zeroing GPS and compass from step 0 collapses all five conditions to 0–6% success (vs. baseline 75–98%): GPS is causally critical across the spectrum. The absolute drops cluster tightly across the four sighted conditions ($\Delta_{\text{coarse}}=-0.940$, $\Delta_{\text{foveated}}=-0.921$, $\Delta_{\text{Fov-LP}}=-0.954$, $\Delta_{\text{uniform}}=-0.972$; blind $\Delta=-0.748$), within a 5 pp band that does *not* track the dissociation direction monotonically. We therefore report this as confirmation

[6] Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K; our values match this null at most pairs.

[7] Scene-id classifier uses scene-stratified 5-fold splits; chance $1/n_{\text{scenes}}$. *Boundaries:* low-resolution (visited-region) occupancy decodes above chance everywhere; metric 20×20 occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025]); ego-frame goal vector is at chance everywhere (PointGoal supplies it as a per-step input). The magnitude and format axes are specifically about the *linearly-decodable, world-frame position code at the policy readout*.

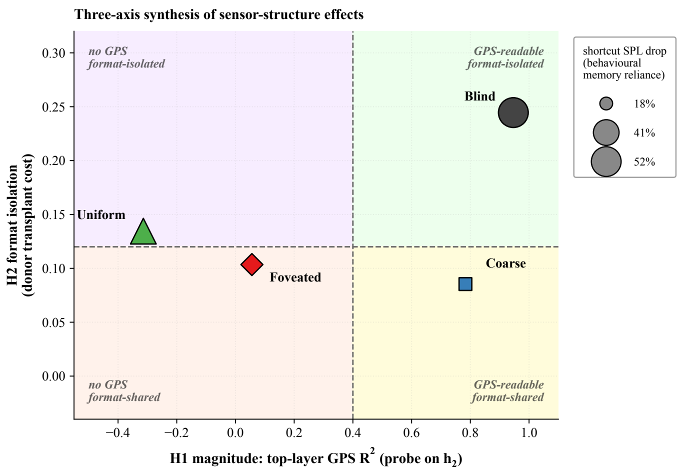

Figure 3: Synthesis: three axes of one capacity allocation. *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (magnitude). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (format-isolation; higher Y = more disruptive to other conditions' policies). *Marker size:* second-episode shortcut SPL drop $\%$ (consumption; larger = more policy dependence on persistent recurrent memory). Within each axis, multiple convergent measurements — e.g. information-allocation, scene-invariance, predictive-horizon, place-cell, and excursion-fragility findings (Figures 25, 22, 27) — sharpen the *interpretation* but are not separate axes of variation.

of GPS causality, *not* as independent ranking-level evidence for the integration–vs–substitute split (which is established at the representational level by the probe and decoder analyses above).[8]

4 Discussion

4.1 Synthesis: three axes of one capacity allocation

The three axes cohere as views of one capacity allocation (Figure 3). *Magnitude* (X): bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§3.2); within-§3.2 measurements (information allocation $\approx 1.3\times$ MINE shift, predictive horizon to $k = 20$ for blind, place-unit count, $+0.381$ MAE excursion fragility) sharpen what the axis means without adding axes. *Format* (Y): within-condition (LOSO scene-invariance dichotomy, Figure 22) and across-condition (asymmetric 5×5 memory transplant + catastrophic cross-condition probe transfer + near-orthogonal subspaces, §3.4) views combine to show subspaces are jointly incompatible at the readout level. *Consumption* (marker size): a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute (scene history) (§3.5). Each condition occupies a distinct (X, Y, size) coordinate, with the bandwidth–allocation tradeoff one of three orthogonal axes, not the whole picture. *Foveation occupies a hybrid position*: it shares uniform's rich-encoder magnitude but separates from uniform on every other measure — most strikingly, foveated agents recover a held-out-MP3D GPS code that uniform agents do not ($R^2 = +0.35$ vs. -0.36, a 0.71-magnitude transferability gap). Foveated–uniform pair similarity ($\theta_1 = 1.08$, kernel-CKA 0.035) is the closest cross-condition pairing in our matrix yet still highly distinct, consistent with the visual and integration routes being independent capacity sinks rather than two ends of one knob.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse's encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform's encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in $\mathbf{h}_2$. The two are not the same mechanism, only two regimes of one allocation.

[8] n=100–300 episodes/sighted; n=50 for blind (no visual sensor $\rightarrow$ every ablated rollout times out at the 2000-step cap, making the full sighted-protocol n wall-clock prohibitive). The mid-rollout visual ablation (a stronger causal test of the magnitude axis) was contaminated by the LSTM going off its training manifold under zero input.

4.2 Biological precedent and predictions

Sensor-niche-shapes-spatial-memory has long-standing precedent in comparative cognition; individual biological observations consistent with it have accumulated over decades. We do not claim our silicon results *demonstrate* the bio-AI parallel, only that the patterns we observe under controlled within-architecture variation are at high level consistent with what one would expect if the same hypothesis held in animals. Conceptually, our LSTM $\mathbf{h}_2$ is most naturally read as a *prefrontal-style activity-slot memory* in the framework of Dorrell et al. [2024]: a cognitive map stored in dynamics rather than synapses, complementary to the synaptic hippocampal map. Two threads are most directly relevant. The bottleneck end of our axis (blind, coarse) parallels animal observations of memory taking over when the visual stream is absent: congenitally blind humans show larger anterior hippocampal volume and supranormal non-visual navigation [Fortin et al., 2008, Kupers and Ptito, 2014], and rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016]. The rich-encoder end parallels the primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — attributed in part to foveated-with-saccades vs. near-panoramic visual sampling [Rolls, 2024]; cross-modality remapping in echolocating bats [Geva-Sagiv et al., 2016] demonstrates sensor-modality-induced shifts in hippocampal coding. Loosely mapped, our *blind* condition is closest to subterranean rodents [Kimchi et al., 2004]; *coarse* to single-channel sensory summaries (echolocation, infrared pit-detection); *foveated* to the primate / raptor regime (gaze fixed; gaze placement is a future-work axis); *uniform* to insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020] — interpretive scaffolding rather than mechanistic identity claims.

Predictions back into biology. Under Sanders et al. [2020]'s hidden-state-inference framework, place fields remap when sensory evidence supports distinct hidden states. Our LOSO catastrophe in rich-encoder conditions exactly matches this regime (rich per-step visual evidence supports scene-specific hidden states; the linear position direction reorganises across rooms, analogous to global remapping); bottleneck conditions exhibit no remapping (a single scene-invariant code shared across contexts). Three falsifiable predictions follow. *(i)* Within-species manipulations changing effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-remapping vs. global-remapping axis. *(ii)* Gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells; Wirth et al. [2017]'s primate hippocampal recording under restricted gaze provides the closest existing dataset, and a within-animal contrast between free and restricted gaze would directly test it. *(iii)* The Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings cohere under one allocation rather than five separate phenomena — cognitive map theory has catalogued such phenomena separately [Tolman, 1948, O'Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], and our setting suggests they may be views of a single mechanism. The structural-constraint reading complements recent results that emergent grid-cell modular structure itself depends on local-interaction priors rather than the path-integration objective alone [Khona et al., 2025, Schaeffer et al., 2022]. Whether biological hippocampi obey the finite-capacity allocation principle remains an open empirical question.

4.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix B), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix D.

4.4 Implications and scope

IB framing (descriptive). The capacity-allocation principle aligns with the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure 22). We use IB descriptively rather than mechanistically; Saxe et al. [2019]'s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, MLP recovery, MINE rank in Appendix E.4) is robust to those concerns.

Falsifiable predictions and broader implications. The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022 for the transformer/hippocampus correspondence), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the same monotonic anti-correlation reported in Figure 2: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator under our 5 sensor conditions; observing the opposite pattern would falsify the principle's architecture-independence and bound it to recurrent-state systems. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity from the downstream substrate's spatial code rather than merely fail to teach it, and bandwidth-restricting perception (foveation, log-polar) may be *cognition-enabling* rather than only compute-saving. The relevant axis is encoder spatial-feature *variety* per step rather than raw input bandwidth (Appendix G); a finer multi-point sweep within recurrent agents is left to future work.

4.5 Limitations

(i) *Single seed per condition*, following the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]. Cross-condition rigour comes from convergent multi-method evidence (linear + MLP probing, CKA, Procrustes, subspace divergence, LOSO, lag-k, 5×5 transplant, Skaggs, excursion-forgetting all rank the conditions in the same direction — unlikely under single-seed noise alone), supplemented by an independent-retraining replication of the blind condition (Appendix A). Per-condition seed-variance error bars over a multi-seed grid are reported as future work. (ii) *Method-family and content-axis scope.* Methods we do not report and that could change a finding's strength: dynamical-systems analysis (fixed-point/slow-point structure [Sussillo and Barak, 2013]; Dynamical Similarity Analysis [Ostrow et al., 2023] as a dynamics-aware Procrustes alternative) would speak to the format axis at the level of computational primitives; information-theoretic estimators of $I(h_t; \mathrm{pos}_{t+k})$ would replace decodability R^2 with predictive-horizon nats; gaze placement and a $\sigma_{\max}$ strength sweep within the foveated condition test the rich-encoder content axis. All four are left for follow-up. (iii) *Architecture, algorithm, and task scope.* We use a recurrent (LSTM) backbone + DD-PPO + PointGoal + ResNet-18 visual encoder for direct comparability with [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. A reasonable concern is whether the encoder–memory race is an LSTM artefact (e.g. vanishing-gradient gating limitations); two within-paper checks argue against it: 2-layer MLP probes recover position in rich-encoder conditions ($R^2 \in [0.51, 0.73]$), so the LSTM *does not* discard position — the linear-readable axis tracks encoder bandwidth, a property of the linear DD-PPO action head; and the cross-condition memory transplant (§3.4) is a behavioural test outside any probe framework that reproduces the same ordering. The encoder–memory race is in principle architecture-agnostic; the cleanest falsification target is a transformer-based navigator under the same conditions, identified in §4 but not verified here. The magnitude- and format-axis patterns could also shift under different RL algorithms, different navigation tasks (ObjectGoal, ImageGoal, exploration), or different visual encoders.

5 Conclusion

A five-condition visual-sensor ablation in a recurrent PointNav agent shows sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle measured along three axes: *magnitude* (bandwidth–allocation tradeoff between encoder and recurrent-integration routes), *format* (within-condition scene-invariance + across-condition non-interchangeable subspaces with asymmetric memory transplant), and *consumption* (probe-readable position dissociates from policy reliance). The bandwidth–allocation ordering is preserved on held-out MP3D, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-/global-remapping axis under hidden-state inference [Sanders et al., 2020] — framed as an open bio–AI hypothesis rather than a demonstrated parallel. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators (the cleanest falsification target, §4). Cross-condition rigour rests on multi-method convergence + an independent-retraining blind replication (Appendix A); a multi-seed grid, finer encoder-bandwidth sweep, and gaze-placement controls are reported as future work.

Code and checkpoints. Code is available at `https://github.com/alunxu/foveated-cog-map`; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §3.2 substitution-dynamics figure) are hosted at `https://huggingface.co/alunxu/spatial-memory-checkpoints`.

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   Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

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A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (`RunningMeanAndVar`) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected; the ablation is in the codebase under `src/habitat/foveated_normalised_policy.py` (normaliser *enabled*) for future work that wants to revisit this choice.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) gradient sanitisation in `wijmans_policy._safe_before_step` replaces NaN/Inf gradients with zero before each PPO update; (ii) a forward-looking `geodesic_distance` clamp in `nan_safe_measures.py` prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories.

Log-polar foveation transform The log-polar control (Appendix G) uses `LogPolarFoveationTransform` in `src/habitat/torch_foveation.py`: resample the 128×128 avg-pooled image onto a 64×64 ($n_\rho \times n_\theta$) log-polar grid before the ResNet-18 backbone. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

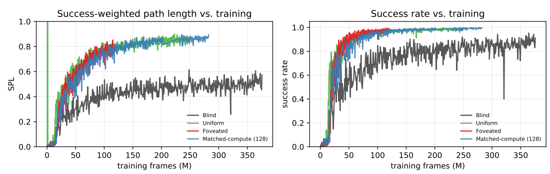
Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

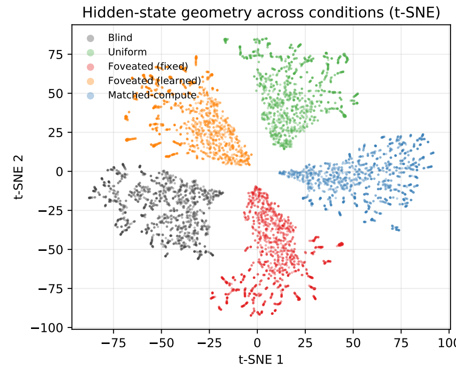
C Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 24c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same



(a) Training curves (TensorBoard event-file scalars). All three sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all four conditions (perplexity 30). The four conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§3.4).

Figure 4: Supplementary visualisations referenced from the main text.

5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge's chance R²). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

D Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), and the real-vs-permuted-label probe gap (Hewitt–Liang selectivity) tracks GPS R^2; the chance-level rich-encoder GPS R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix C). H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

E Supplementary visualisations

E.1 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. 25) and LOSO scene-invariance (Fig. 22) results in §3.2.

E.2 Single-direction β-scrubbing reveals redundant population coding

The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space

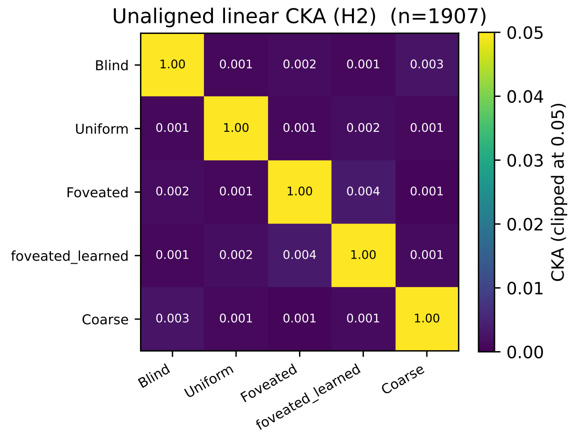


Figure 5: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 23); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

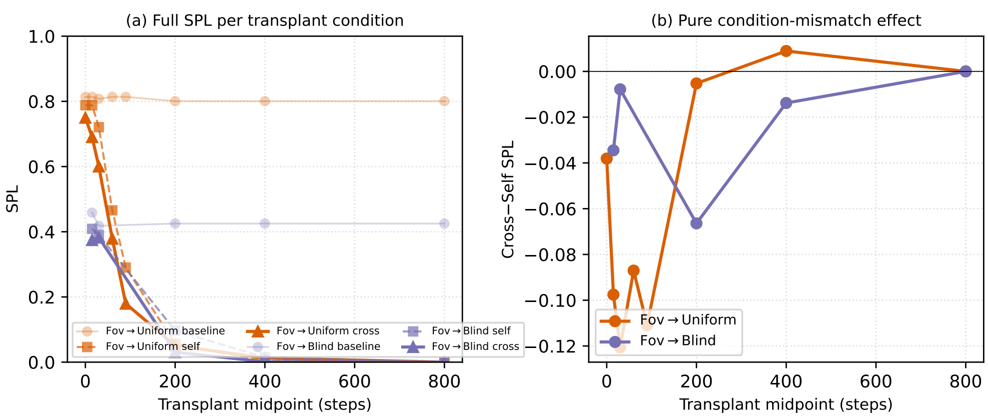


Figure 6: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in Figure 23 (right) is at midpoint t=30. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (so the diagonal in Figure 23 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 in Figure 23 is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	+0.95	+0.93	+0.94	+0.95
Coarse	+0.72	+0.62	+0.79	+0.84
Uniform	−1.08	−1.46	+0.42	+0.31

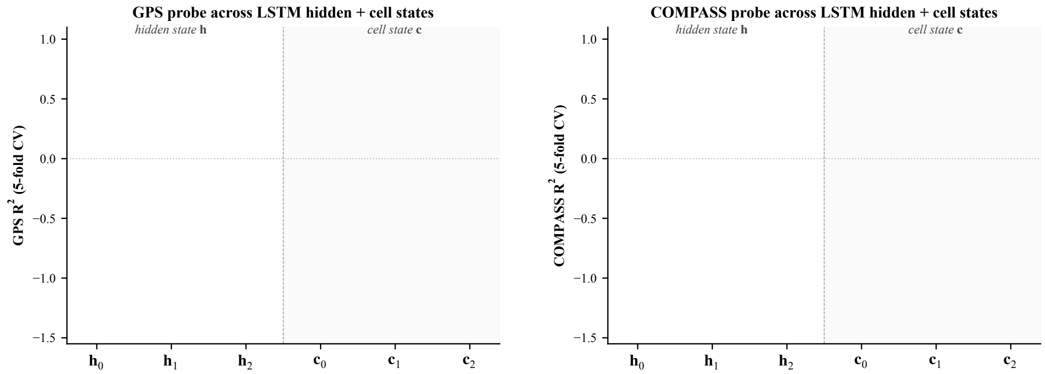


Figure 7: Linear probe (R^2, 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about "top-layer linearly-decodable GPS" is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

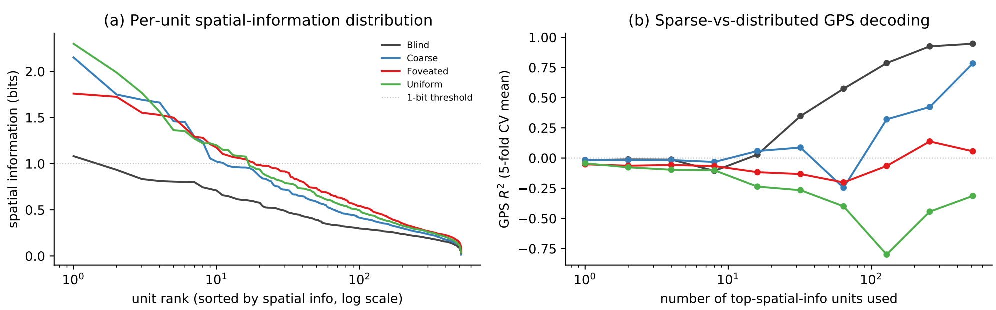


Figure 8: Population coding analysis (referenced from §**??**). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max $\approx$ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information* $\neq$ *position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank

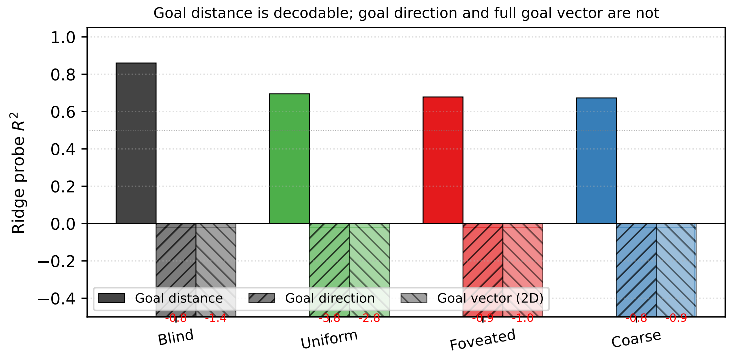


Figure 9: Goal-vector probe R^2 (referenced from §**??**). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

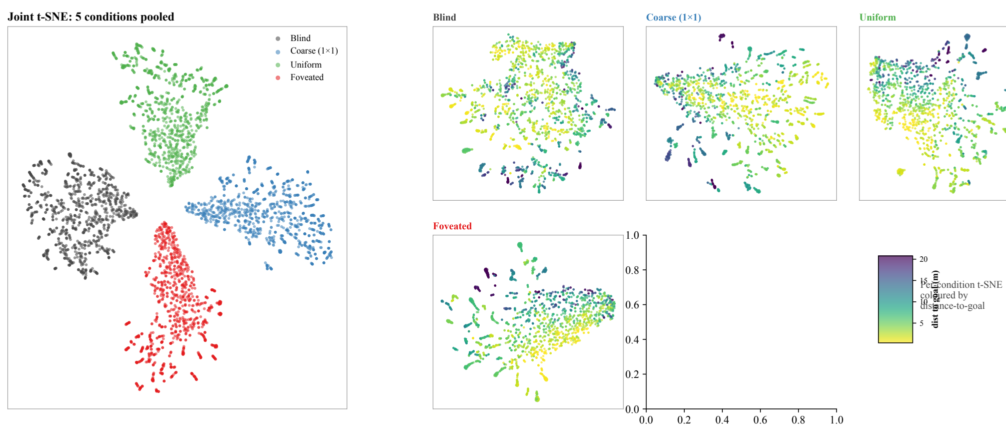


Figure 10: t-SNE of top-layer LSTM hidden states $\mathbf{h}_2$, supplementing the linear-similarity H2 results (§3.4). *Left:* joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

E.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear or MLP probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

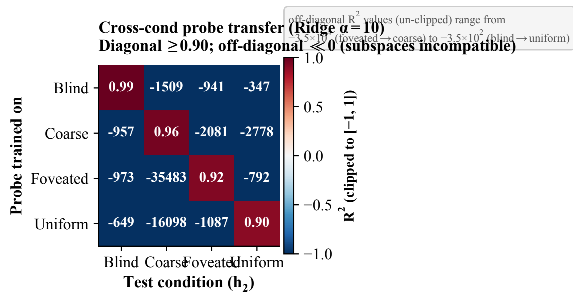


Figure 11: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A's top-layer states, applied to condition B's. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* ($-3.5{\times}10^2$ to $-3.5{\times}10^4$ unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 28) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

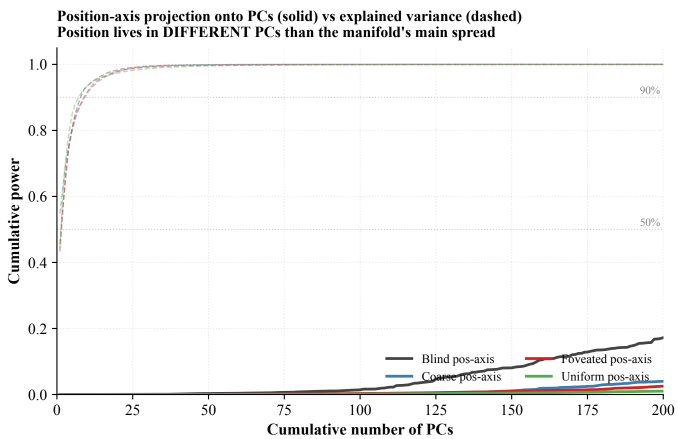


Figure 12: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 22).

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
Linear (Ridge $\alpha = 10$) R^2, all-episode pooling, episode-level 5-fold CV						
Blind	$+0.94$	$+0.94$	$+0.94$	$+0.92$	$+0.90$	$+0.79$
Coarse	$+0.57$	$+0.57$	$+0.57$	$+0.56$	$+0.50$	$+0.37$
Foveated	$+0.06$	$+0.06$	$+0.08$	$+0.13$	$+0.19$	$+0.06$
Foveated-logpolar	$+0.17$	$+0.18$	$+0.17$	$+0.11$	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains ≥ 0.90 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.79 at $k = 50$. That is, the same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.94$ also decodes positions 20 steps in the future at $R^2 = 0.90$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k = 0$–20, dropping to $+0.37$ at $k = 50$).

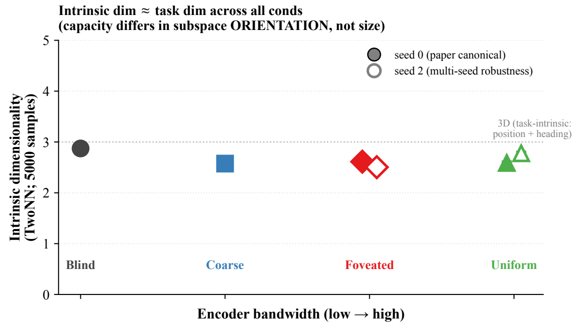


Figure 13: **Intrinsic dimensionality (TwoNN).** Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5–2.9, comparable to the task-intrinsic dimensionality (2-D position + 1-D heading $\approx$ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 28), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

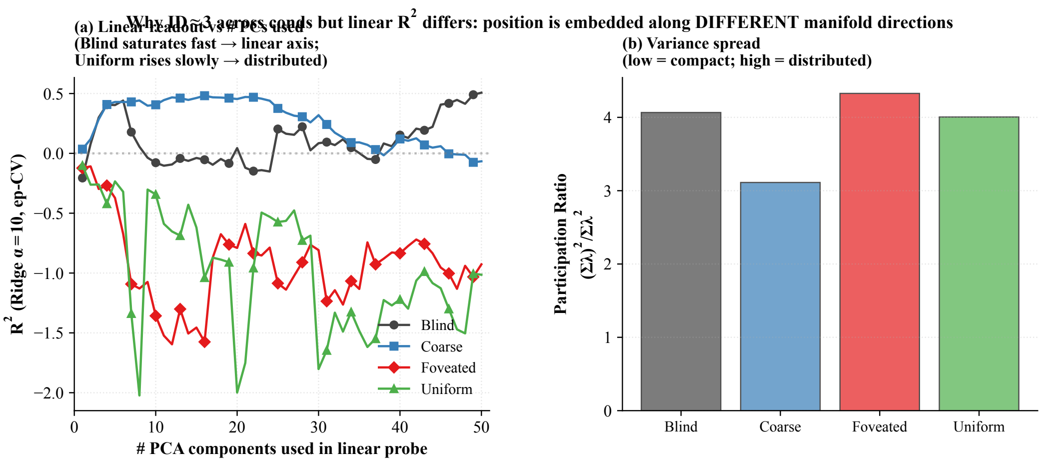


Figure 14: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 12).

(iii) Rich-encoder conditions have no clear predictive horizon. Foveated stays in $[0.06, 0.19]$ across $k = 0$–50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that rich-encoder position codes are scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

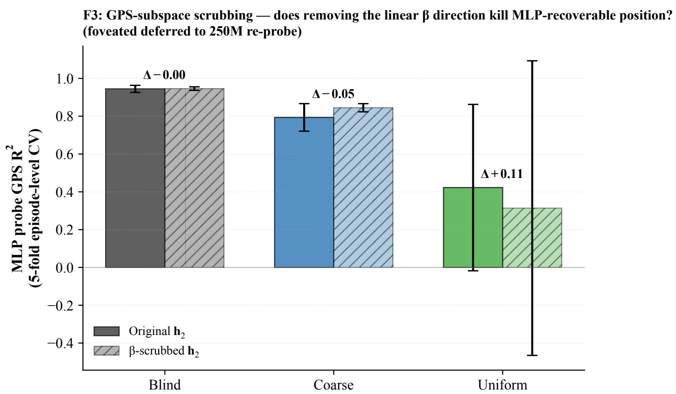


Figure 15: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^+\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

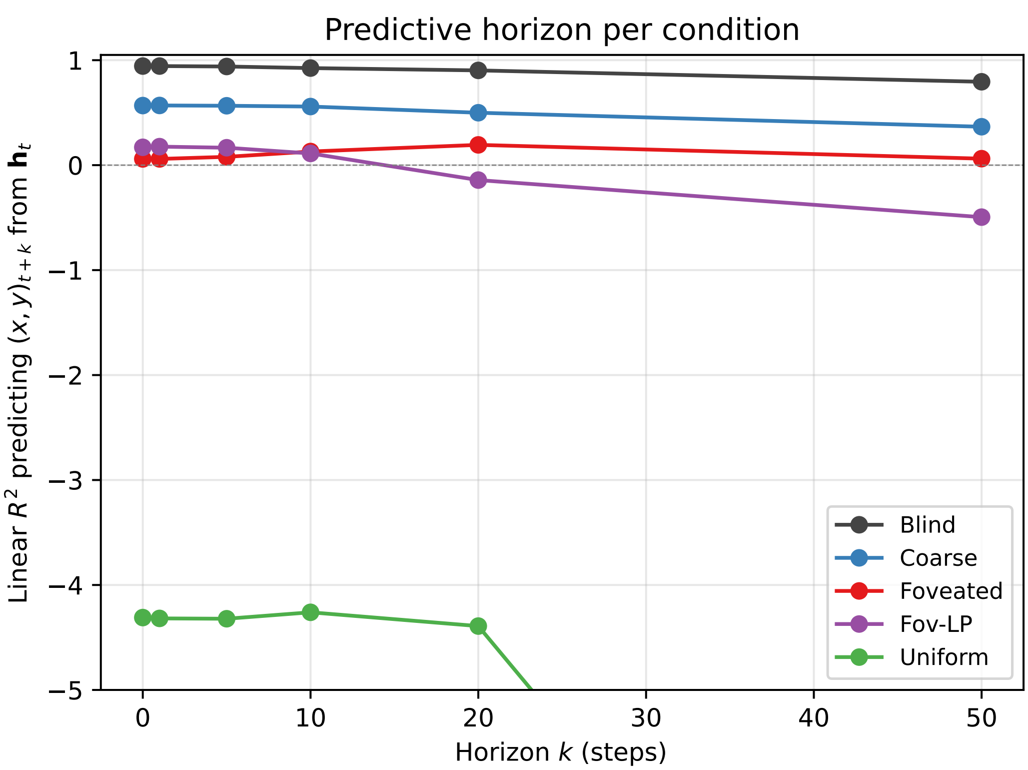


Figure 16: **Predictive horizon: linear** R^2 **predicting** $(x, y)_{t+k}$ **from** h_t **as a function of horizon** k**.** Blind retains a long predictive horizon (linear R^2 stays ≥ 0.90 to $k = 20$, $+0.79$ at $k = 50$); Coarse a moderate one (linear $R^2 \in [0.50, 0.57]$ to $k = 20$, $+0.37$ at $k = 50$); rich-encoder conditions have no predictive horizon: foveated stays in $[0.06, 0.19]$ with no monotonic structure, foveated-logpolar starts $+0.17$ then turns negative by $k = 20$, and uniform is strongly negative at $k = 0$ already (-4.31) and grows more negative. 500 episodes per condition (full pool), mean-centred $\mathbf{h}_t$, episode-level 5-fold CV, episode boundaries respected (target $(x, y)_{t+k}$ taken from the same episode as the input $\mathbf{h}_t$).

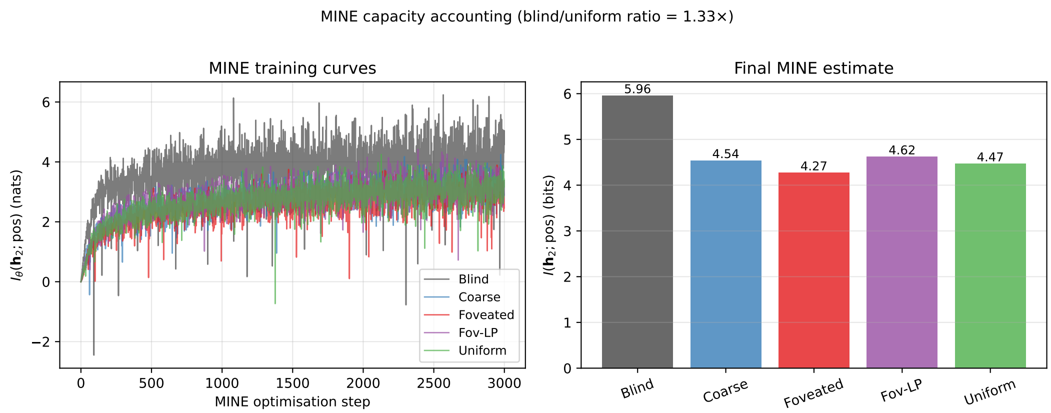


Figure 17: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim 3{,}000$ steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06, Coarse 4.55 ± 0.03, Foveated 4.30 ± 0.04, Foveated-logpolar 4.66 ± 0.04, Uniform 4.47 ± 0.06. Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std $\leq$ 0.06 bits, much smaller than the within-sighted gap $\approx$ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 22).

E.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §3.2 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \text{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X; Z) \geq \sup_T \mathbb{E}_{P_{XZ}}[T(x, z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x, z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \text{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

E.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 18, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading), and

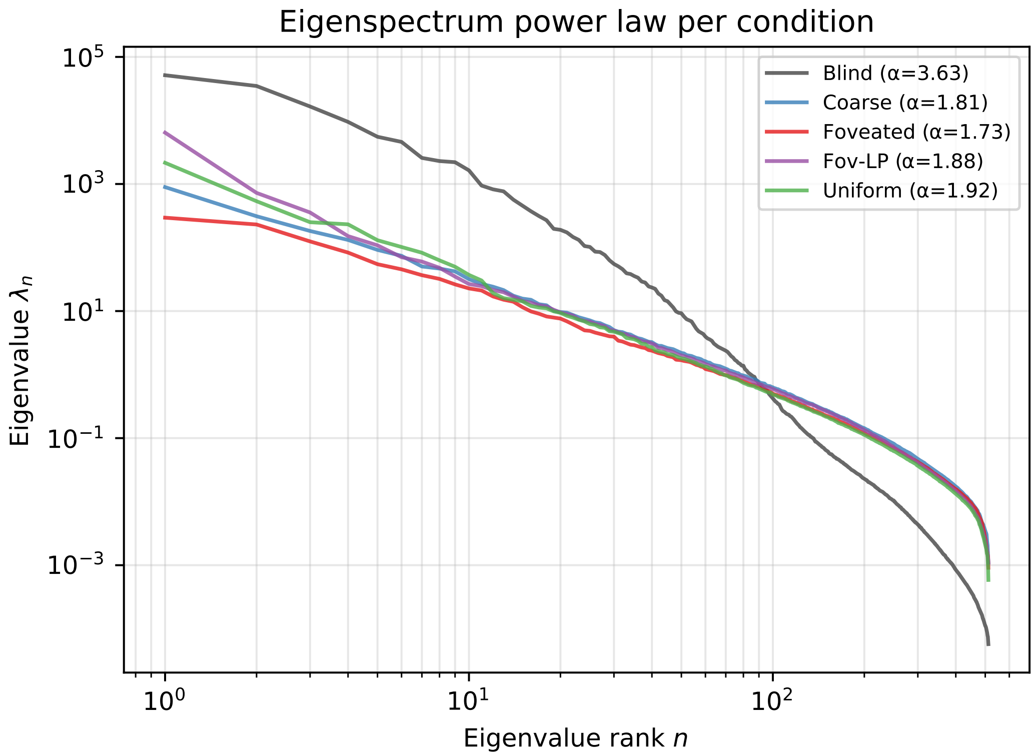


Figure 18: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

far above Stringer's V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind's eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx 1.7$–1.9, all near the $1 + 2/d_{\text{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim concentrated regime, while the four sighted conditions distribute variance across more PCs. The split bottleneck-vs-rich-encoder reading: blind's heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose 1×1 encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

E.6 Persistent-memory failure-mode catalog

The main-text Figure 30 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 19) shows four cases per condition spanning the per-condition margin range (most-tries-new $\to$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$ (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

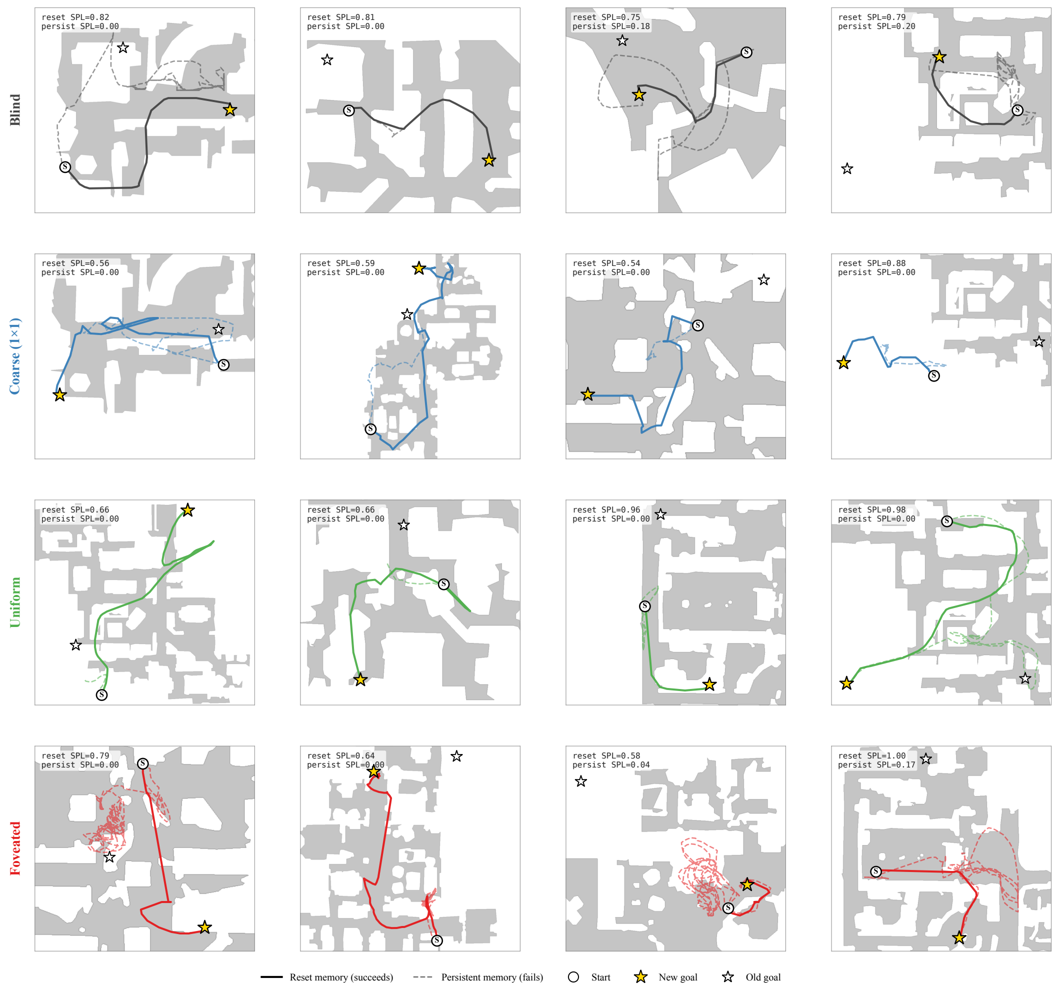
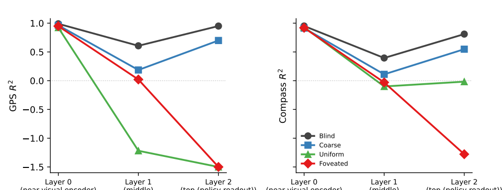
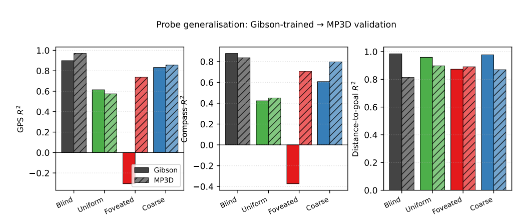


Figure 19: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see §4.5).



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 24a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-rich-encoder partition is the same.

Figure 20: Per-layer and MP3D companion figures (referenced from §3.2).

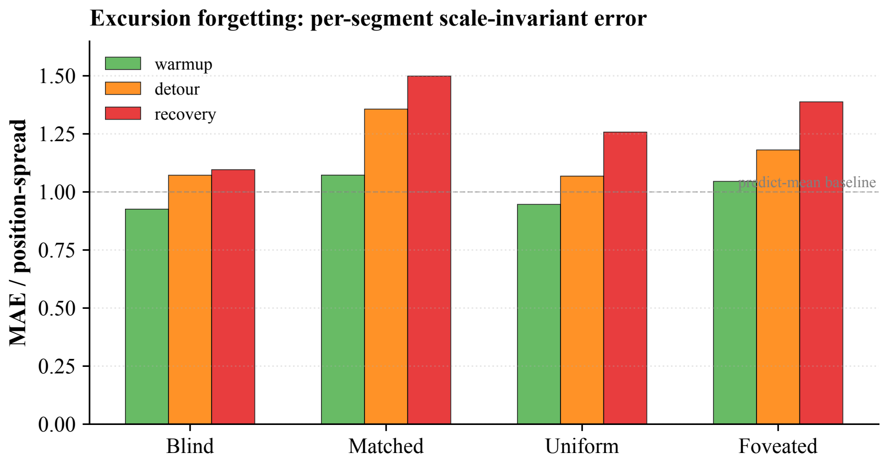


Figure 21: Excursion-forgetting per-segment scale-invariant error (referenced from §**??**). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All five conditions show recovery $>$ warmup error (forgetting); blind's rise ($+0.381$) is the largest, roughly 1.8–$2.8\times$ the sighted conditions ($+0.135$ foveated-logpolar, $+0.167$ uniform, $+0.198$ coarse, $+0.210$ foveated). The pattern is consistent with blind carrying the most action-stream-dependent integrated GPS code and the sighted conditions partly re-anchoring on visual landmarks during recovery. Single seed.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see §4**

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from the main paper. All four are submitted.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists

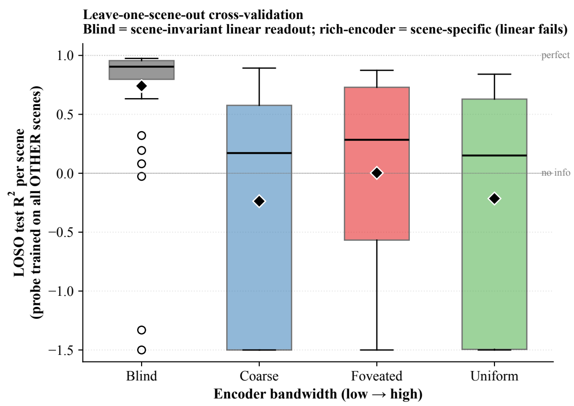


Figure 22: **Leave-one-scene-out (LOSO) cross-validation.** For each cond, Ridge probe trained on ~ 30 Gibson scenes (top scenes by sample count) and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: $38-48\%$ of held-out scenes give negative R^2. Encoder bandwidth dictates whether the linear position-axis is scene-invariant (linearly probable across scenes) or scene-conditional (requires non-linear readout to switch contexts).

vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2\times2$, intermediate between coarse's 1×1 and uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another axis. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, $\sim$ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 26) corroborates this reading: log-polar starts at $R^2 = +0.92$ at $\sim$ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at $\sim$ 250M frames each) was outside this submission's compute window.

H Deferred main-text figures

The figures below are referenced from main-text §3 but moved here to keep main text within the page budget. Each is a quantitative companion to the corresponding paragraph; the body text reports the headline numerical claims directly.

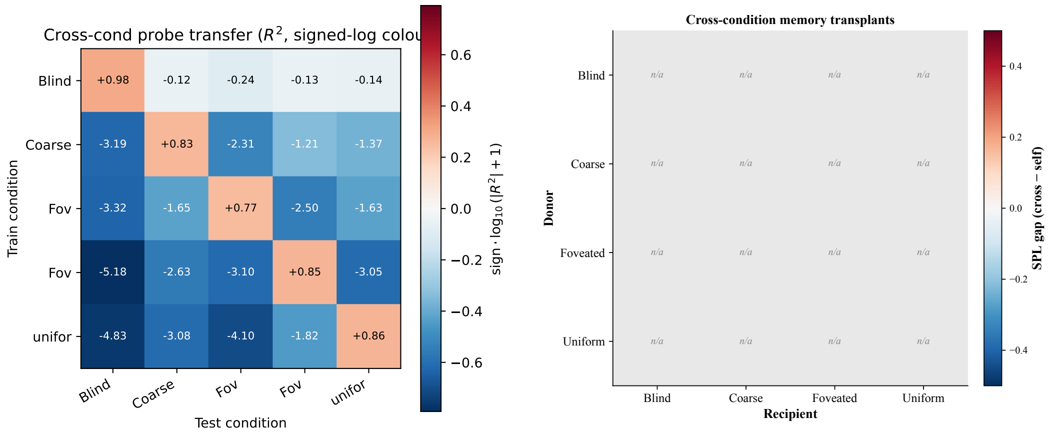


Figure 23: Format-axis (across-condition) evidence, two views. *Left:* cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *Right:* 5×5 cross-condition memory-transplant matrix at midpoint $t{=}30$; off-diagonal = cross-transplant SPL $-$ recipient's self-transplant SPL (6 coarse cross-pair cells excluded due to differing visual input size). Single seed; midpoint-sweep companion in Appendix E.

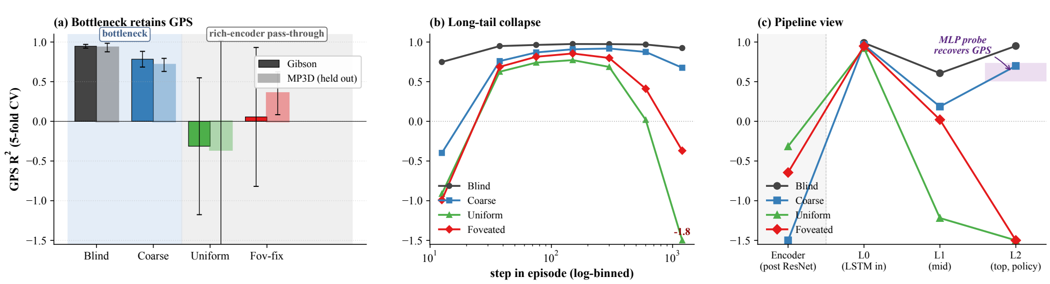


Figure 24: Magnitude-axis evidence in detail. *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). *(b)* Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024). *(c)* GPS R^2 along Encoder $\rightarrow$ L0 (sensor concat) $\rightarrow$ L1 $\rightarrow$ L2 (policy-readable). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$). Blind has no encoder.

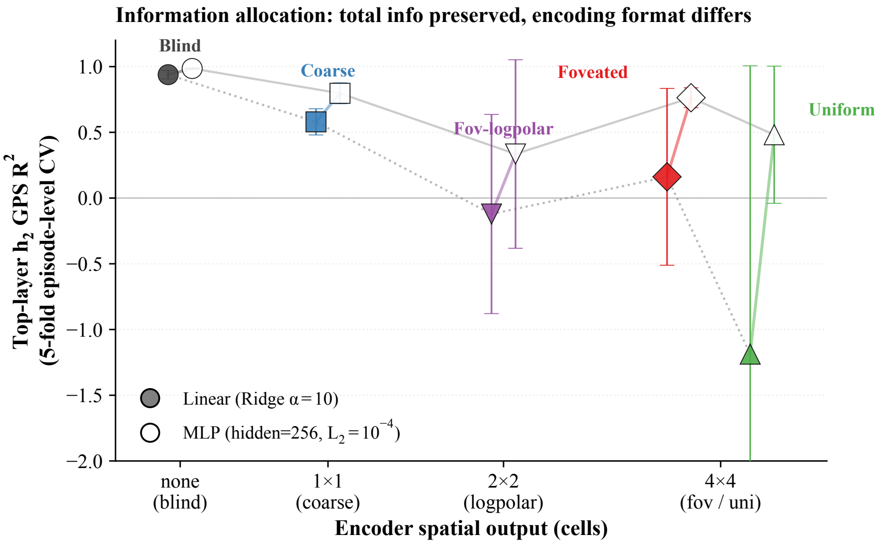


Figure 25: **Information allocation: linear vs. MLP readout.** Per-condition top-layer $\mathbf{h}_2$ GPS R^2 under linear (filled markers, dotted line) and 2-layer MLP probing (hollow markers, solid line); same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5. Position information is approximately preserved by MLP probing in rich-encoder conditions even when linear readability collapses — a format shift, not a total-information collapse. MINE-estimated total $I(\mathbf{h}_2; \text{pos})$ shows a more modest $\approx 1.3\times$ decrease across conditions (Appendix E.4).

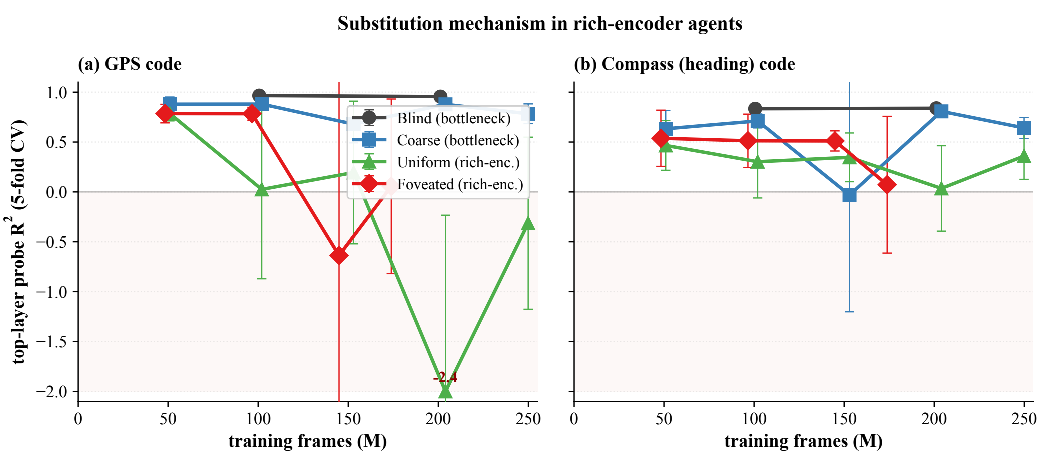


Figure 26: Cross-training probe profile across 4–5 training checkpoints per condition, truncated at 250M frames. *(a)* Top-layer GPS R^2. *(b)* Top-layer compass R^2. (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative-R^2 outliers clipped at -2.0.)

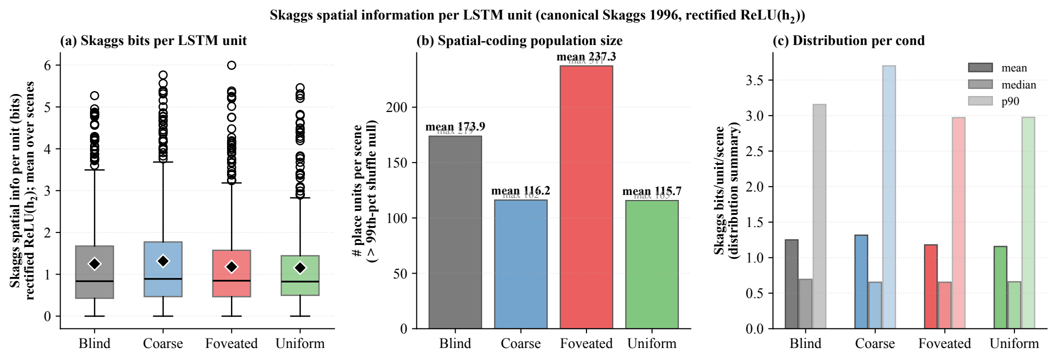


Figure 27: **Skaggs spatial information per LSTM unit.** *(a)* Distribution of per-unit Skaggs bits (averaged across 15 top scenes per condition); bottleneck and rich-encoder conditions overlap substantially (means $\approx$ 1.16–1.32 bits). *(b)* Number of place units per scene exceeding the 99th percentile of a 100-shuffle null; foveated has the largest count ($\approx$ 237/scene), blind/coarse/uniform $\approx$ 116–174. (c) Distribution summary; cross-condition differences sit within the within-condition spread. The canonical Skaggs metric on rectified activations does not by itself discriminate bottleneck from rich-encoder regimes.

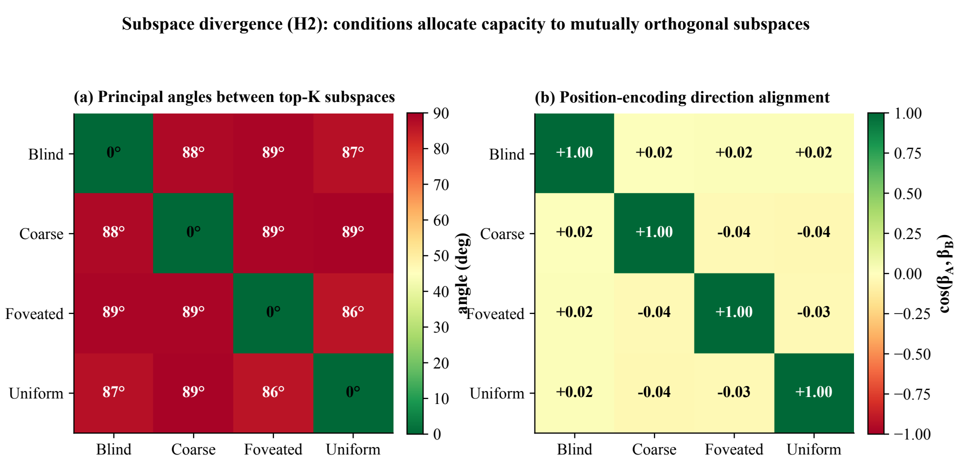


Figure 28: **Subspace divergence, quantitative form.** *(a)* Mean principal angles between top-K PCA subspaces of $\mathbf{h}_2$ ($K \in [4, 33]$ per condition, covering $\sim 90\%$ variance; 20 000 episode-step samples per condition). Off-diagonal angles in $[69, 87]$ (most pairs in $[77, 87]$) — conditions occupy mostly-orthogonal subspaces; the closest pair (foveated–foveated-logpolar, 69) is the same-family rich-encoder pair. *(b)* Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.094, +0.102]$ — position is encoded along essentially uncorrelated directions across conditions. Single seed.

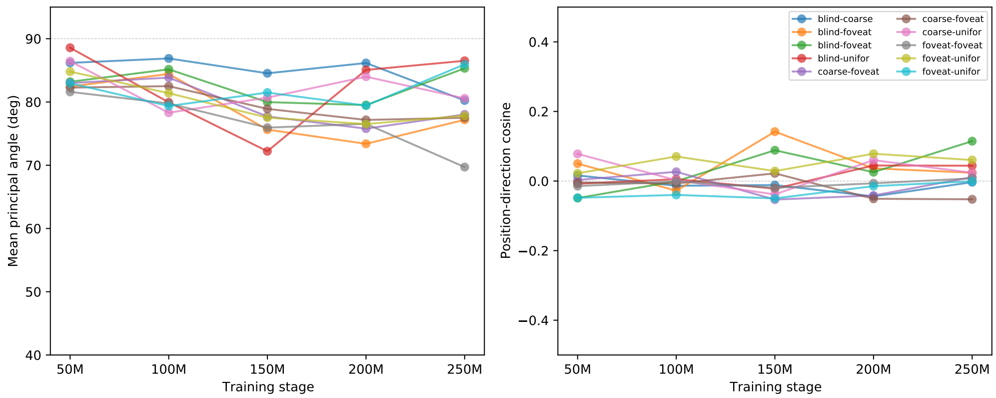


Figure 29: **Subspace evolution over training.** *(a)* Mean principal angle between condition-pair top-K subspaces, plotted across five training stages ($\sim$ 50M to $\sim$ 250M frames; sighted at ckpt 10/20/30/40/49, blind at 10/15/20/25). Angles are already $[82, 89]$ at the earliest probed stage and stay near-orthogonal throughout. *(b)* Position-encoding direction cosines are essentially zero throughout (range $[-0.05, +0.14]$), indicating direction divergence is committed before the earliest probed checkpoint. Single seed; 4–5 ckpts per condition.

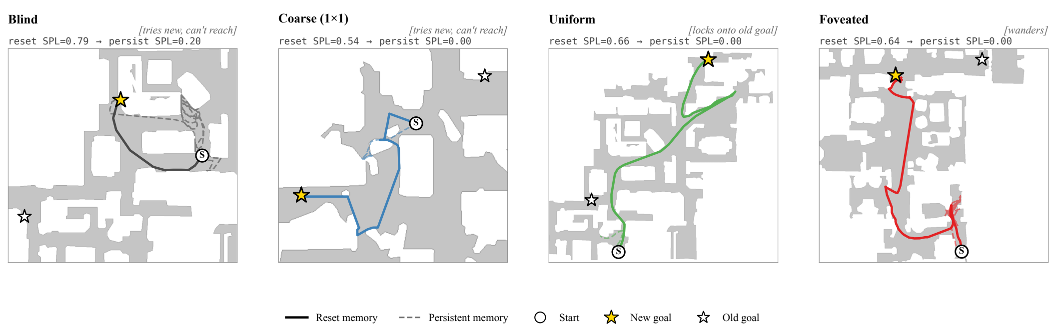


Figure 30: Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix E.6). Aggregate per-condition margin = avg dist(persistent end $\rightarrow$ old goal) $-$ avg dist(persistent end $\rightarrow$ new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Foveated -0.59m (n=16), **Uniform $+1.83$m (n=46, closer to OLD goal)**. Single seed.